IDV2025 - Final Report

Introduction

Finnish roads still claim about ~100 lives and injure roughly 2 000 people each year, despite a slow downward trend [1], [2]. The national Road-Safety Strategy 2022–2026 emphasises data-driven action for safer infrastructure and better-informed road users [3].

We publish an open Tableau dashboard that turns a decade of accident records into an interactive map-and-chart exploration tool for *policy-makers, analysts, and everyday drivers.* It lets users

- spot spatial hot-spots and high-risk road segments,
- compare seasonal and long-term trends (2014–2023), and
- relate crash counts to severity, traffic volume, and weather-road conditions.

By coupling coordinated views with intuitive filters, the visualisation supports evidence-based decisions and raises public awareness of when and where driving risks are highest.

Dataset and Data Abstraction

We use the Väylävirasto dataset on Finnish road traffic accidents from 2014 to 2023 [4]. It comprises two tables per year: one with detailed accident records (our primary data source) and another containing information about involved parties (*osalliset*). Additionally, road shape data used for selecting roads comes from Väylävirasto's Tieosoiteverkko.

Preprocessing

Due to changes in data formatting between 2021 and 2022, we harmonized variable names and standardized reporting practices across all years. This preprocessing step included:

- Standardizing representations of missing values, replacing various terms (e.g., 'Ei tietoa', 'Ei arvoa', '-') uniformly with `NaN`.
- Normalizing weekday notations (e.g., standardizing 'maanantai' and 'ma').
- Converting geographical coordinates to standardized longitude and latitude (WGS84).
- Enhancing accident records by including maximum vehicle age and weight information from the *osalliset* table.
- Assigning road classes (`tieluokka`) based on road numbers according to the Finnish road numbering classification.

Selected Variables

We used the following variables from the dataset: `onnettomuus_id`, `tie`, `maakunta`, `kunta`, `vuosi`, `kuukausi`, `viikonpäivä`, `vakavuus`, `osallisten_lkm`, `tietyö`, `onnettomuuspaikka`, `liittymäjärjestelyt`, `pinta`, `valoisuus`, `lämpötila`, `sää`, `nopeusrajoitus`, `päällyste`, `rautatie`, `kvl`, `kvl_raskas`, `näkemä%_yli_150`, `näkemä%_yli_300`, `näkemä%_yli_450`, `pvm`, `lon`, `lat`, `ajoneuvon_massa_max`, `ajoneuvon_ikä_max`, `kuolemaan_johtanut`, `tieluokka`

Additionally, we derived two metrics to enrich the analysis:

- **Fatality rate**: Proportion of fatal accidents to total accidents.
- **Risk**: Number of accidents normalized by annual average daily traffic (`kvl`).

Variable Categorization

We categorized variables as follows:

Numerical:
Temperature (`lämpötila`), visibility distances (`näkemä%_yli_150`, `näkemä%_yli_300`, `näkemä%_yli_450`), average daily traffic (`kvl`, `kvl_raskas`), speed limit (`nopeusrajoitus`), maximum vehicle weight (`ajoneuvon_massa_max`), maximum vehicle age (`ajoneuvon_ikä_max`).

Ordinal:
Accident severity (`vakavuus`), road surface condition (`pinta`), brightness (`valoisuus`), day of the week (`viikonpäivä`), road class (`tieluokka`).

Nominal:
Weather (`sää`), road surface type (`päällyste`), municipality (`kunta`), region (`maakunta`), junction type (`liittymäjärjestelyt`), railway involvement (`rautatie`), roadworks (`tietyö`).

Task Abstraction

Table 1 outlines the key user tasks in accordance with Munzner's nested model for visualization design [5], and draws on Brehmer and Munzner's multi-level typology of abstract visualization tasks [6], which frames tasks in terms of *what* the user is doing, *why* they are doing it, and *how* it is supported.

ID	What?	Why?	How?
T1	Gain an *overview* of accident distribution and frequency across months and years	Detect seasonality and long-term trends for high-level planning	Line charts and scatter plots; focus-and-context colouring (red highlights chosen year); hover tooltips reveal exact values; highlights related series from other plots.
T2	*Examine* detailed spatial patterns of accidents	Spot geographic hotspots and inspect severity to prioritise interventions	Interactive map with zoom/pan; circles coloured by severity and sized by persons involved; density mini-maps
T3	*Compare* accident characteristics across years and conditions (weather, temperature, road surface, . . .)	Understand contributing factors and generate hypotheses	Scatter-matrix; click column header to highlight matching points on the map; log-scale toggle exposes subtle spikes
T4	*Relate* accident counts to estimated traffic volume (compute *risk*)	Normalise raw counts so lightly-trafficked roads aren't undervalued	Derived metric overlay in line charts; a dedicated risk density mini-map
T5	*Identify* anomalies and trends in fatality rate and risk over time	Discover outlier years that deviate from expected safety progress	Interactive year filter; risk/fatality rows in time-series view;
T6	*Analyze* one road's multi-year accident history	Support case-study analysis for specific road segments	Filter by road number (select road via text box or map); all views update; tool-tips show full accident record

Table 1: Tasks mapped to *What–Why–How*, following @behmer2013

Design Rationale

Table 2 summarizes the different views, their visual encodings and interactions, and the tasks they support. The four views together provide detailed spatial and temporal insights, aligned with the tasks defined earlier. The choice to include multiple views stems from the multifaceted nature of the dataset: spatial relationships are best explored using maps, while temporal patterns are more clearly represented through charts. In addition, global filters were included to allow users to narrow down the data to specific years, road types, or individual road IDs—both to improve performance (e.g., reducing map clutter and rendering delays) and to help users focus on relevant subsets.

View	Encodings & interactions	Related tasks
Main map	• Circle position = lat/lon • Color (3-level) = severity • Size = persons involved • Hover tooltip → full accident record • Severity highlighting • Pan / zoom	T2, T3, T4, T6
Mini-maps	• Density heat-maps for *Accidents*, *Fatality Rate*, *Risk*	T2, T4
Time-series view	• X = selected variable; Y=metric (3 stacked rows for *Accidents*, *Fatality Rate*, *Risk*) • Columns = year (or all in one panel) • Focus+context color: red = selected year, grey = others • Hover highlights same year in scatter-matrix	T1, T3, T4, T5
Scatter-matrix	• X = year/month, Y = metric • Columns = variable bins • Linear/Log-scale selector • Selecting a bin filters maps • Hover highlights same year in timeseries	T3, T4, T5
Global controls	• Year slider / buttons • Road-class check box • Road-ID text field filter or mini selector map	T1–T6 (all)

Table 2: Views, their visual encodings and interactions, and the tasks they support

When selecting visual encodings, we referred to the principles of Mackinlay's expressiveness and effectiveness criteria for graphical representation design [7]. For example, accident severity is encoded using color hue and saturation—channels that are particularly effective for distinguishing nominal and ordinal variables. Grey was used for the lowest severity (no injuries), orange for injury-level accidents, and red for fatal ones. The traditional green color was deliberately excluded to avoid any unintended positive connotations. This choice aligns with the recommendations of Moreland [8], who cautions against overly broad or misleading color spectrums that can introduce perceptual ambiguity.

Although size is generally considered a less effective channel for comparison [7], we chose to use it on the map to represent the number of people involved in each accident. While subtle differences may be hard to distinguish, this encoding helps highlight exceptionally large accidents. To manage overlap, we used hollow circles rather than filled dots, enabling visibility of overlapping incidents.

The time-series and scatter plots were inspired by Edward Tufte's work [9], [10]. While Tableau limits some aspects of minimalist design, we strived to follow Tufte's principles by reducing non-data ink and using subdued greys for context. The use of red to highlight the selected year in time-series charts also mirrors his approach of visually elevating key data without cluttering the view.

We aimed to make full use of Tableau's built-in interactive capabilities, enabling dynamic filtering and highlighting across multiple views. For example, hovering over a year in one chart automatically highlights the corresponding data in other views, facilitating easier comparison and analysis. Additionally, the ability to switch between different data series allows users to explore the full potential of the dataset while conserving screen space.

Iterative Design Process

The design process began with a period of exploratory data analysis. We familiarized ourselves with the dataset by experimenting with different variables and verifying whether the data behaved as expected. The first prototype consisted of a simple map visualizing major roads and accident locations, created using Python (see Figure 1). At this early stage, Python was used both for data preprocessing and visualization.

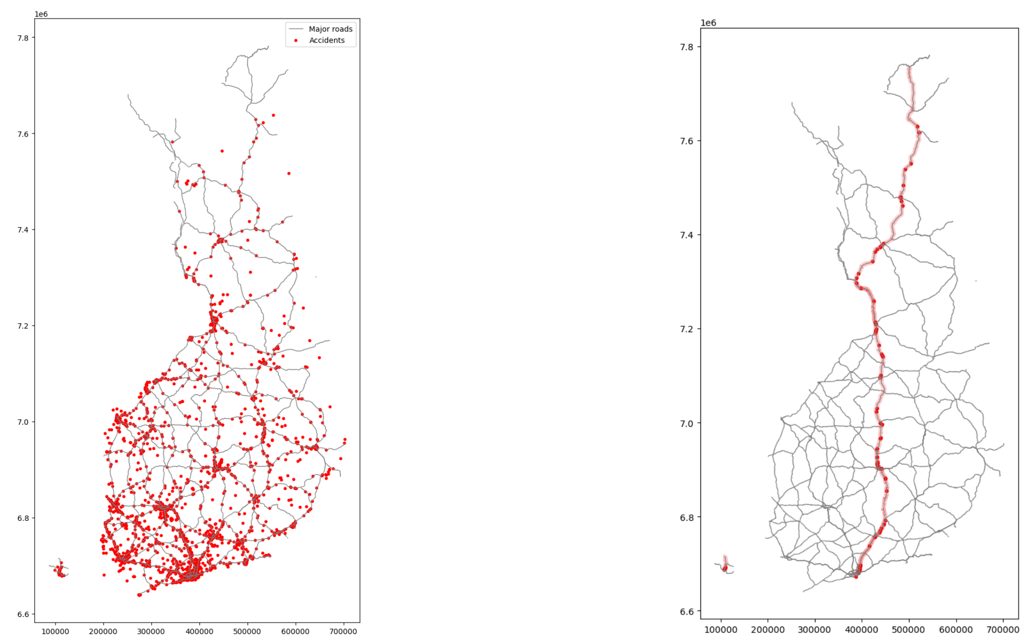

Figure 1: Early map prototypes: accidents shown on all major roads (left) and filtered to a single road (right).

To better understand relationships between variables, we next implemented an interactive correlation matrix using Plotly (Figure 2). Although this revealed some interesting patterns, the approach was ultimately abandoned in favor of focusing more directly on accidents, their severity, and derived risk metrics. We also explored histograms of individual variables (Figure 3), which proved useful primarily for validating preprocessing steps rather than for final visualization design.

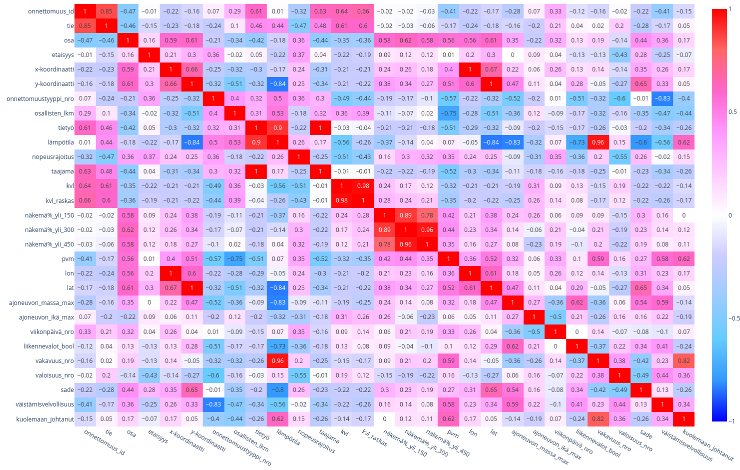

Figure 2: Interactive correlation matrix created with Plotly, used to explore relationships between variables in the dataset during early analysis.

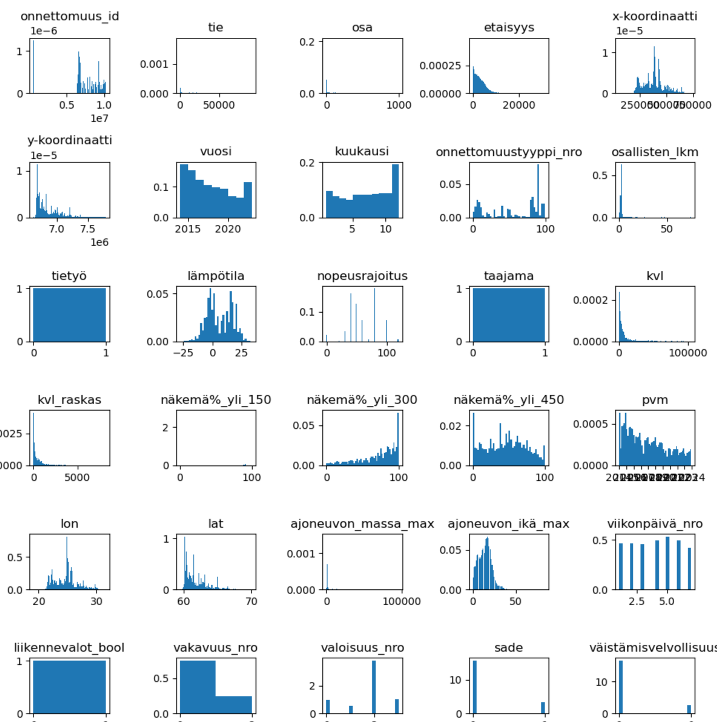

Figure 3: Histograms of individual variables, used primarily to validate data preprocessing and check for expected distributions.

Recognizing the complexity and multidimensionality of the data, we introduced more structure to the design process by applying the Five Design-Sheet Methodology [11]. Due to time constraints, we completed only the first two sheets, but these already provided significant clarity. At this stage, we were not yet constrained by Tableau Public's performance limitations—especially the latency of its web interface—which allowed for more conceptual freedom in the initial design phase. The sketches from this step are shown in Figure 4.

Figure 4: Early conceptual sketches based on the Five Design-Sheet Methodology: Sheet 1 (brain-storming) and Sheet 2 (initial desings).

Eventually, we transitioned to Tableau as the primary visualization platform. This decision was driven by the desire to avoid manual UI programming and to gain familiarity with a widely-used industry tool. However, performance was an issue, as our Linux environment only supported the web-based interface, which was noticeably slow during development.

An initial Tableau prototype was built and used in a user evaluation study (Figure 5). Based on feedback, we refined the layout, expanded interactive functionality, and adjusted visual encodings. For instance, an earlier green–yellow color scheme in the time-series view was found to be ambiguous and was subsequently revised. Several incremental improvements followed—though not detailed here—culminating in the finalized version shown in Figure 6.

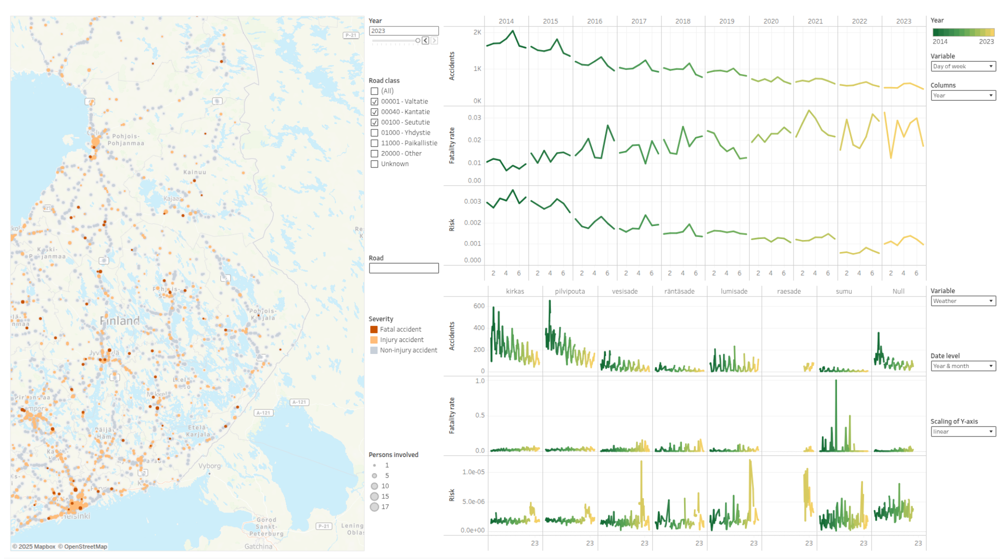

Figure 5: Initial Tableau prototype used for user evaluation. Early color and layout choices were revised based on feedback.

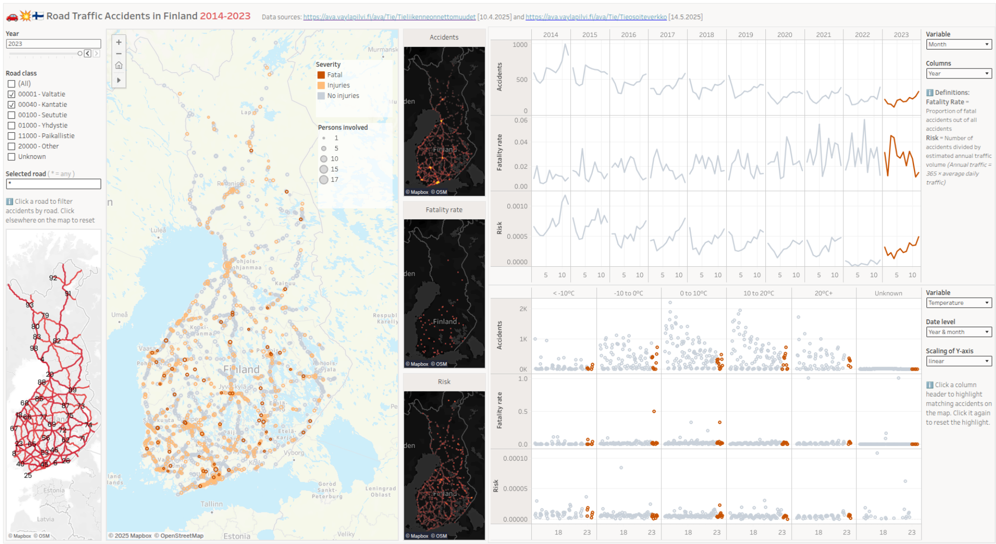

Figure 6: Final version of the interactive dashboard, featuring refined layout, improved color encoding, and enhanced interactivity.

Lessons Learned

- *Tool limitations matter.* While good design does not depend on tools alone, they can significantly affect the process. Working on Linux without access to Tableau Desktop meant relying on the slower web-based version, which introduced interface lag, limited functionality, and ultimately restricted further exploration. In a professional setting, it's important to ensure your technical setup supports efficient workflow from the start.
- *Preprocessing is essential groundwork.* Although data cleaning and preprocessing can feel tedious, they are critical for successful visualization. That said, this step can be integrated into an iterative process—starting with only the fields required for initial views and refining the pipeline as needed.
- *Thoughtful interactivity requires iteration.* Designing meaningful interactions—where each operation is both valid and justified—demands careful thought and multiple iterations. It is not enough to simply enable interaction; the experience must serve analytical goals.
- *Structured design aids clarity.* Following a structured design methodology, such as the Five Design-Sheet approach, helped organize ideas and prioritize features. Creating initial sketches on paper was particularly effective in encouraging exploration and saving time, compared to experimenting directly in a slower digital interface.

References

[1] Statistics Finland, "Road traffic accidents [online database]." 2024. Available: https://stat.fi/tilasto/ton

[2] Liikenneturva, "Henkilövahingot henkilöautossa." 2024. Available: https://www.liikenneturva.fi/tutkimukset/henkilovahingot-henkiloautossa/

[3] F. M. of Transport and Communications, "Liikenneturvallisuusstrategia 2022–2026." 2022. Available: https://julkaisut.valtioneuvosto.fi/handle/10024/163951

[4] Väylävirasto, "Väylävirasto - aineiston välitysalusta." 2025. Available: https://ava.vaylapilvi.fi/ava/Tie/Tieliikenneonnettomuudet

[5] T. Munzner, "A nested model for visualization design and validation," *IEEE transactions on visualization and computer graphics*, vol. 15, no. 6, pp. 921–928, 2009.

[6] M. Brehmer and T. Munzner, “A multi-level typology of abstract visualization tasks,” *IEEE transactions on visualization and computer graphics*, vol. 19, no. 12, pp. 2376–2385, 2013.

[7] J. Mackinlay, “Automating the design of graphical presentations of relational information,” *ACM Trans. Graph.*, vol. 5, no. 2, pp. 110–141, Apr. 1986, doi: 10.1145/22949.22950.

[8] K. Moreland, “Diverging color maps for scientific visualization (expanded),” *Proceedings in ISVC*, vol. 9, pp. 1–20, 2024.

[9] E. R. Tufte, *The visual display of quantitative information*, Second. Cheshire, Connecticut: Graphics Press, 2001.

[10] E. R. Tufte, *Beautiful evidence*. Cheshire, Connecticut: Graphics Press, 2010.

[11] J. C. Roberts, C. Headleand, and P. D. Ritsos, “Sketching designs using the five design-sheet methodology,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 22, no. 1, pp. 419–428, 2016, doi: 10.1109/TVCG.2015.2467271.